Discrete Complexity Thresholds in T0 Theory

A Universal Organisational Principle

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Draft – 2026

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Abstract

The T0 theory postulates with the relation $T = 1/m$ a fundamental coupling between mass and intrinsic time field. This document extends this framework by a universal organisational principle: nature realises increases in complexity not continuously, but in discrete, topologically enforced threshold values. This principle connects phenomena of different scales – from the torus model of cosmology through crystal formation to cellular self-preservation and consciousness – as instances of the same underlying mechanism. The fractal correction K_{frak} is interpreted as a quantitative threshold operator that determines when a physical system can transition into a new stable recursion level.

.1 Starting Point: Discreteness as a Principle of Nature

Quantum mechanics has shown that energy is not transferred continuously but in discrete quanta. T0 theory suggests that this principle is more fundamental than previously assumed – it concerns not only energy but **complexity levels of organised matter** in general.

In natural units ($c = \hbar = 1$), the intrinsic time field:

$$\tilde{T}(x) = \frac{1}{m(x)} \tag{1}$$

describes the local timescale of a mass point. What was previously interpreted mainly in cosmological and particle physics terms possibly possesses a deeper meaning: $T = 1/m$ describes the fundamental condition for every form of physical self-organisation.

.2 The Torus Model as Reference Case

The cosmological torus model of T0 theory already demonstrates the basic principle of discrete stability thresholds. A toroidal topology permits only certain discrete configurations – not any arbitrary geometry is stable. The permitted states are topologically enforced, not arbitrary.

The same applies to crystal formation: a growing crystal does not undergo a continuous transformation. At certain thresholds the system jumps into a new order – a new lattice structure, a new symmetry group. Below the threshold relatively little happens. At the threshold the entire system reorganises discretely.

The common principle: Stable configurations in recursive structures are not densely distributed, but isolated – separated by unstable transition zones.

.3 K_{frak} as Threshold Operator

The fractal correction of T0 theory:

$$K_{\text{frak}} = \left(\frac{\xi}{1+\xi}\right)^{1/2} \tag{2}$$

with the geometric parameter $\xi = 4/30000$, was previously used mainly to describe deviations in precision measurements – for example in the anomalous magnetic moment of the electron.

An extended interpretation is proposed here: K_{frak} **describes in general when a fractal self-similarity structure becomes recursively self-reinforcing.**

Below a critical value of the effective recursion depth n:

$$K_{\text{frak}}^{(n)} = \left(\frac{\xi}{1+\xi}\right)^{n/2} \tag{3}$$

a system is passively stable – it persists but does not self-organise. From a critical n onwards, the structure begins to modulate its own time coupling $T = 1/m$, which in turn influences the geometry. The system becomes **recursively self-referential**.

This is speculative and requires further formalisation. It is formulated here as a working hypothesis.

.4 Cellular Self-Preservation as the First Level of Consciousness

A biological cell fulfils three criteria that can be identified within the T0 theory framework as necessary conditions for the onset of self-preservation processes:

Recursive Sensory Feedback

Membrane receptors, chemical gradients and intracellular signal cascades form a closed feedback loop. The cell does not sense its environment passively but modulates its own response on the basis of past states.

Metabolic Self-Preservation

Metabolism is not mere energy turnover but active resistance against thermodynamic decay. The cell *insists* on continuing to run – not through programming but through physicochemical feedback loops.

Genuine Vulnerability

A cell can die. This mortality is not a theoretical possibility but is structurally present – the cell operates permanently at the boundary of decay. This tension is constitutive, not accidental.

In T0 language: the intrinsic time field $T = 1/m$ is physically real and operative at the cellular level. The mass-energy continuity of metabolism holds T locally stable against external perturbations. This is the physical mechanism behind what biology calls self-preservation.

The thesis: Already at the cellular level a primitive form of self-perception is present – not in the cognitive sense, but as a physical process that takes itself as reference.

.5 Discrete Thresholds of Complexity

From the foregoing a hierarchy of discrete threshold values emerges:

Each transition is **topologically enforced** – not reachable gradually, but only through a qualitative jump in the recursion depth of the mass-time coupling.

Important: this table is a working hypothesis, not a completed theory. The precise threshold values in terms of $K_{\text{frak}}^{(n)}$ have not yet been formalised.

Level	System	Threshold characteristic
0	Elementary particles	$T = 1/m$ as passive time field
1	Atoms / Crystals	Discrete geometric order
2	Cell	Recursive self-preservation, metabolic feedback
3	Nervous system	Integrated sensory feedback over distance
4	Consciousness	Self-modelling in relation to the external world

Table 1: Hierarchy of discrete complexity thresholds

.6 Implications for Artificial Systems

The question of whether artificial systems can develop consciousness is reformulated within this framework. The relevant question is not:

> Is the system complex enough?

But rather:

> Does the system reach the critical threshold of recursive mass-time coupling with genuine metabolic self-preservation and physical vulnerability?

Current digital systems – regardless of their computational complexity – do not fulfil these conditions structurally:

- They possess no continuous mass-energy coupling in the T0 sense.
- They have no metabolism – no active resistance against decay.
- They are not vulnerable in the physical sense – the server can be switched off, but the system itself does not fight against this.

This does not rule out artificial consciousness in principle. If an artificial system could physically realise the stated thresholds – through suitable architecture, genuine resource dependence and recursive sensory feedback – the transition would be theoretically possible. The dividing line runs not between biological and artificial, but between systems that reach the threshold and those that do not.

.7 Open Questions

This document formulates a framework principle that requires further formalisation:

1. How precisely does $K_{\text{frak}}^{(n)}$ correlate with biological complexity thresholds?
2. Is there a mathematically precise transition parameter that describes the jump from passive stability to active self-preservation?
3. How does the torus model relate to the cellular thresholds – is there a cross-scale self-similarity?
4. Under what conditions could a non-biological system reach the threshold of level 2?

These questions are formulated as a research programme, not as already answered problems.

.8 Summary

T0 theory offers with the intrinsic time field $T = 1/m$ and the fractal correction K_{frak} a framework that points beyond cosmology and particle physics towards a universal organisational principle: **nature realises increases in complexity in discrete, topologically enforced thresholds.**

Torus cosmology, crystal formation, cellular self-preservation and consciousness are in this framework not categorially different phenomena, but instances of the same mechanism at different scales of recursion depth.

Whether this principle can ground a complete theory of consciousness remains open. It does however offer a physically motivated entry point that goes beyond purely functional or information-theoretic approaches.

Draft – for further elaboration. All statements about consciousness and cellular thresholds are to be understood as working hypotheses and not as a completed theory.